Traditional Rag

Embeddings : converting text into vectors

documents → Vector DB → Cloud / if it is local → Vector Store

Documents refer to sentence or query that we ask

embedding is the mathematical representation of any sort of document

SQL NOSQL VECTOR DATABASE

Vector store

Set of numbers (array)

data → Chunking → Embedding → Vector database

Similarity search
(2d distance,cosine similarity)

User → i/p → LLM → Relevant data → User

Vector Database

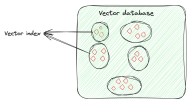
1. In memory database → Own server
2. on disk database
3. cloud database → AWS/google/azure/pinecone/weaviate/quadrant

IN-Memory
on disk → RAG → Gen AI

1. index
2. similarity search

pdf → Pages → Document Objects

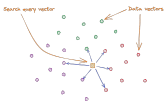
Chunking



Types of indexes

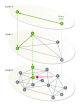
1. Flat → Exact Match
2. HNS
3. IVF → Approximate match

Flat Index



flat index goes each and every node because it is a exact match

HNSW and IVF



In hnsw and ivf it is not going to search all the nodes

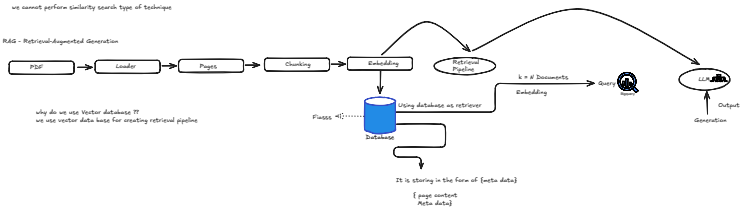
no sql → { json schema }
sql → Rigid Schema
Vector DB → RAG → Retrieval Pipeline

why not we use other databases than vector db??

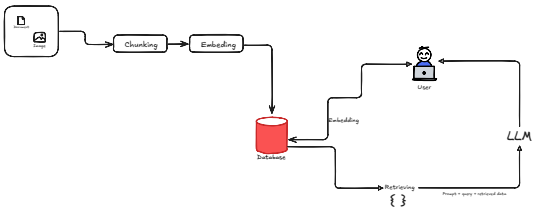
In both no sql and sql vector operations is not possible

we cannot perform similarity search type of technique

RAG - Retrieval-Augmented Generation



RAG ARCHITECTURE



local and in memory → FIASS

Cloud Host → Pine cone
Quadrant and milvus
weaviate cloud

Flat index → EXACT Match → Fetching k = n documents

HNSW → Graph method → Fetching k = n documents

IVF → Cluster method → Fetching k = n documents